

HADI KARIMI

PhD Candidate in Geoscience (Earth Science / Remote Sensing)
Terrestrial Water Cycle | Earth Observation | Land-Atmospheric Coupling | Climate Extremes
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EDUCATION

2022 - PRESENT

PhD Candidate in Geoscience (Earth Science/Remote Sensing), WESTERN MICHIGAN UNIVERSITY

Department of Geological and Environmental Sciences

Research & Teaching Assistant

2012 - 2015

M.S. in Geological Remote Sensing, AZAD UNIVERSITY OF ISFAHAN (KHORASGAN)

2006 - 2011

B.S. in Geology, AZAD UNIVERSITY OF ISFAHAN (KHORASGAN)

ACADEMIC APPOINTMENTS

2024 - PRESENT

RESEARCH ASSISTANT EARTH SCIENCE REMOTE SENSING FACILITY

WESTERN MICHIGAN UNIVERSITY, Kalamazoo, MI, USA

- Led development of ML/DL modeling in the lab and remote-sensing derived datasets (GRACE/GRACE-FO, IMERG, Sentinel/Landsat/Planet) for geospatial analysis, hydrologic studies and model evaluation.
- Built reproducible geospatial pipelines to preprocess/validate flood and water-storage signals; generated publication-quality figures and grant reports.
- Contributed to proposal development and reporting for climate change and geospatial research grants (ESRS).
- Collaborated with external partners over the grant (e.g., Saudi Geological Survey / KSU) to deliver interpreted outputs and documentation (more info <https://www.esrs.wmich.edu/>)

2022 - 2024

TEACHING ASSISTANT, WESTERN MICHIGAN UNIVERSITY

Geological and environmental Science Department, Kalamazoo, MI, USA

(Teaching Assistant (Lab):

- Taught lab sections in Remote Sensing / GIS / UAVs; mentored student projects, grading, and technical troubleshooting.
- Developed lab materials/tutorials supporting geospatial analysis workflows.

RESEARCH EXPERIENCE

2025

PHYSICS-AWARE TERRESTRIAL WATER STORAGE FORECASTING USING A CLIMATE FOUNDATION MODEL (ClimaX foundation model | Microsoft)

(PROJECT IN FINAL STAGE)

- Fine-tuning ClimaX, a Vision Transformer-based climate foundation model, to forecast GRACE/GRACE-FO terrestrial water storage anomalies at three-month lead time over semi-arid basins, with emphasis on physically interpretable subseasonal-to-seasonal predictability.
- Designed strict out-of-sample validation over the 2020–2025 Great Basin drought period, including persistence and trend baselines, multi-seed ensembles, spatial NSE/bias/correlation maps, and per-year regime-shift diagnostics.
- Developed a per-pixel trend-deflation strategy to preserve spatial hydroclimatic structure in pretrained transformer forecasts, improving drought-period skill and reducing negative-skill spatial hotspots.

2025

(CLIMATE EXTREMES AND FLOOD MODELING)

- Implemented 2D hydrodynamic flood modeling and validated simulations using Sentinel-2 and PlanetScope observations for ungauged basins.
- Applied EVT with bootstrap confidence intervals to quantify flood probability shifts and uncertainty across spatiotemporal datasets.
- Used satellite and model-based constraints to evaluate how extreme precipitation, runoff response, and terrestrial water-storage anomalies interact in arid land-surface systems.
- Contributed to model validation and applied bias correction workflow using GPM IMERG, and GRACE-FO datasets.

2025

(ASSESSMENT OF SABKHA AND DUNE DYNAMICS IN MECCA AND RIYADH PROVINCES)

- Participating in a collaborative project between the Saudi Geological Survey and KSU's Earth Sciences Remote Sensing team to assess sabkha and dune dynamics in Mecca and Riyadh Provinces (2023-2025), contributing to data processing and development, interpretation, and report writing.

2024

(DEEP LEARNING FOR LARGE-SCALE SAND DUNE MAPPING (SEMANTIC SEGMENTATION))

- Applied convolutional neural networks (U-Net variants) to high-resolution PlanetScope and multispectral Sentinel-2/Landsat imagery for dune identification over the Arabian Peninsula.
- Achieved >92% IoU, validated across >5,000 test images, enabling automated and scalable monitoring of geomorphologic change in arid environments (GPU training and implementation).

2024

(MACHINE LEARNING FOR GROUNDWATER QUALITY MAPPING)

- Develop stacking ensemble machine learning framework for groundwater-quality prediction (QDA + AdaBoost meta-learner); reached >0.90 accuracy (ROC≈0.96) and outperformed individual base learners

2023

(MACHINE LEARNING OPTIMIZATION FOR GROUNDWATER QUALITY MODELING)

- Applied Artificial Neural Networks (ANN) integrated with Particle Swarm Optimization (PSO) and Grey Wolf Optimizer (GWO) to model and map groundwater quality across an unconfined aquifer in Northern Iran.

2023

Water Storage and Hydrologic Systems Analysis in Arid Regions; Utilizing GRACE and GRACE-FO TWS Solutions

2022

Evaluation of Impact of extreme precipitation event on terrestrial water storage

2013 - 2014

(MASTER RESEARCH TRAINING)

- Study and analysis of different correction methods such as atmospheric and topographic corrections for classification and mapping remote areas
- Determining the process of spatial distribution of Arsenic and Proven and Relativity of Arsenic with geological distribution
- Evaluation in application of remote sensing methods in estimating and evaluating the amount of land subsidence
- Evaluation of spatial resolution and spectral resolution in different sensors and their fields of usage

PEER-REVIEWED PUBLICATIONS

2025

Elhaddad, H., Sultan, M., Yan, E., Tran, D., Torres-Urbe, H.E. and Karimi, H., 2025. Nile basin flow regimes under 21st century climate variability. *Communications Earth & Environment*, 6(1), p.880.

2025

Karimi, H., Sultan, M., Eugene, Y., Elhaddad, H., Saleh, H., Abdelmohsen, K., Emil, M.K., "Climate-extreme modeling framework for sustainable flood management in the Arabian Peninsula." *Journal of Environmental Management* 393 (2025).
<https://doi.org/10.1016/j.jenvman.2025.127074>

2025

Saleh, H., Sultan, M., Yan, E., Save, H., Elhaddad, H., Karimi, H., Abdelmohsen, K., Emil, M.K., Al Qamshouai, S. Intensifying tropical cyclones in the Arabian Sea replenish depleting aquifers. *Commun Earth Environ* 6, 536 (2025).
<https://doi.org/10.1038/s43247-025-02493-w>

2025

Karimi, H., Sahour, S., Khanbeyki, M. et al. Enhancing groundwater quality prediction through ensemble machine learning techniques. *Environ Monit Assess* 197, 21 (2025). <https://doi.org/10.1007/s10661-024-13506-0>

2025

Mohammad AT, Sultan M, Abotalib AZ, Voice P, Saleh H, Karimi H, Emil MK, Elhaddad H. Coupled control of tectonic and surface processes on the inception and evolution of the East Saharan mega-depressions. *Geomorphology*. 2025 Sep 17:110025.

2024

Elhaddad, H., Sultan, M., Yan, E., Abdelmohsen, K., Mohammad, A.T., Badawy, A., Karimi, H., Saleh, H. and Emil, M.K., 2024. Optimization of floodwater redistribution from Lake Nasser could recharge Egypt's aquifers and mitigate its excessive floods. *Communications Earth & Environment*, 5(1), p.385. <https://doi.org/10.1038/s43247-024-01532-2>

2023

Sahour, S., Khanbeyki, M., Gholami, V., Sahour, H., Karimi, H., & Mohammadi, M. (2023). Particle swarm and grey wolf optimization: enhancing groundwater quality models through artificial neural networks. *Stochastic Environmental Research and Risk Assessment*, 1-15.

2023

Sahour, S., Khanbeyki, M., Gholami, V., Sahour, H., Kahvazade, I. and Karimi, H., 2023. "Evaluation of machine learning algorithms for groundwater quality modeling". Environmental Science and Pollution Research Journal, pp.1-18.

2022

Shahabi-Ghahfarokhi, S., Rahmati-Abkenar, M., Matson, J.G., Karimi, H., Yu, C., Hogland, W., Klavins, M. and Ketzer, M., 2022. "Removal and potential recovery of dissolved metals from acid sulfate soil drainage by spent coffee grounds and dissolved organic carbon". Environmental Advances Journal, 8, p.100193.

AWARDS AND HONORS

2026

Gwen Frostic Doctoral Dissertation Fellowship at Western Michigan University

2025

Climate Change Research Graduate Grant at Western Michigan University

2024

Climate Change Research Graduate Grant at Western Michigan University

2023

Geological Society of America (GSA) Graduate Student Research Grant

2022

Scholarship Awarded at the Department of Geological and Environmental Science at Western Michigan University

2013

Top-ranked Student at Department of Geology, Azad university of Isfahan, Iran

CONFERENCE ABSTRACTS AND PRESENTATIONS

2024

Karimi, H., Sultan, M., Elhaddad, H., Emil, M.K., & Abdelmohsen, K. (2025). Data-driven framework for quantifying dryland water storage patterns and near-future projections. AGU Fall Meeting.

2024

Karimi, H., Sultan, M., Emil, M.K., & Elhaddad, H. (2024). Deep learning for large-scale sand dune mapping: A convolutional neural network approach. AGU Fall Meeting, Abstract EP51D-1373.

2024

Karimi, H., Sultan, M., Yan, E., Abdelmohsen, K., Emil, M.K., Elhaddad, H., & Saleh, H. (2024). Utilizing extreme precipitation to mitigate water scarcity in arid areas: A hydrological approach for natural and artificial recharge. AGU Fall Meeting, Abstract H53D-1132..

2023

Karimi, H., Sultan, M., Abdelmohsen, K., Emil, M.K., Yan, E., Saleh, H., & Othman, A. (2023). The impact of extreme precipitation events on terrestrial water storage in the Arabian Peninsula. AGU Fall Meeting.

2022

Karimi, H., Sultan, M., Emil, M.K., Sahour, H., Abdelmohsen, K., & Saleh, H. (2022). The impact of extreme precipitation events on terrestrial water storage in the Arabian Peninsula. AGU Fall Meeting.

2022

Saleh, H., Sultan, M., Abdelmohsen, K., Save, H., Karimi, H. and Emil, M.K., 2022, December. "Use of GRACE and GRACE-FO to Monitor the Impacts of Tropical Cyclones on Arabia's Hydrologic Systems." In Fall Meeting 2022. AGU Conference (Abstract)

2022

Sultan, M., Karimi, H., Saleh, H., & Save, H. (2022). GRACE response to climate change over engineered and nonengineered basins. Copernicus Meetings, GSTM2022-70.

MANUSCRIPTS UNDER REVIEW

2026

Karimi, H., Sultan, M., Emil, M.K., Elhaddad, H., " Adaptive Resilience in Mobile Landscapes: Defining Upper-Bound Risk Envelopes for Future Compound Climate Extremes" (*Manuscript under review at Journal of Environmental Management.*)

SKILLS

Climate and Earth Observation Data: GRACE/GRACE-FO, GPM IMERG, ERA5/ERA5-Land, NLDAS, CMIP6/NEX-GDDP, Sentinel-2, Landsat, PlanetScope

Programming and Scientific Computing: Python, PyTorch, xarray, rioarray, NumPy, pandas, GeoPandas, Dask, GDAL, scikit-learn

Machine Learning and AI: Vision Transformers, ClimaX, U-Net/UNet++, ensemble learning, feature engineering, uncertainty analysis, SHAP

Geospatial and Remote Sensing: ArcGIS Pro, QGIS, Google Earth Engine, ENVI, raster/vector processing, hydrologic basin analysis

Statistics and Climate Extremes: EVT, GEV, bootstrap uncertainty, bias correction, quantile delta mapping, time-series analysis

Reproducibility and HPC: Git, Docker, CUDA GPUs, DDP/NCCL, mixed-precision training

PROFESSIONAL SERVICE (PEER REVIEW)

2026

- Reviewed for Nature Portfolio Scientific Report

2025

- Reviewed for Applied Soft Computing Journal
- Reviewed for Water Resource Research Journal
- Reviewed for Environmental Earth Sciences Journal
- Reviewed for Environmental Monitoring and Assessment Journal

2024

- Reviewed for Environmental Processes Journal

2023

- Reviewed for Arabian Journal of Geosciences
- Reviewed for Frontiers (Environmental Informatics and Remote Sensing Journal)